

## Homework 3

**LU factorization with partial pivoting** Let  $A \in \mathbb{R}^{n \times n}$ , and suppose that Gaussian elimination with partial pivoting produces the factorization

$$PA = LU,$$

where  $P$  is a permutation matrix,  $L$  is unit lower triangular, and  $U$  is upper triangular.

1. Prove that the multipliers used in Gaussian elimination with partial pivoting satisfy

$$|\ell_{ij}| \leq 1, \quad i > j.$$

2. Let

$$\|A\|_{\infty} := \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|$$

denote the induced infinity matrix norm. Prove that

$$\|U\|_{\infty} \leq 2^{n-1} \|A\|_{\infty}.$$

*Hint:* Let  $A^{(k)}$  denote the matrix at the beginning of elimination step  $k$ . Show that, because

$$|\ell_{ik}| \leq 1,$$

the row sums of the matrix can increase by at most a factor of 2 at each elimination step. Use the fact that row permutations do not change the infinity norm.

3. Define the element growth factor by

$$\rho_n := \frac{\max_{i,j} |u_{ij}|}{\max_{i,j} |a_{ij}|}.$$

Show that

$$\rho_n \leq 2^{n-1}.$$

*Hint:* Part (2) does not follow from part (3), and part (3) does not follow from part (2). You have to do independent proofs for both. If you go from (2) to (3), you get an  $n$  term and this is not good. If you go from (3) to (2), you get an  $n$  term and this is not good. So do not try to prove (3) from (2). Both (2) and (3) are useful results.

4. Consider the matrix

$$W_n = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & 1 \\ -1 & 1 & 0 & \cdots & 0 & 1 \\ -1 & -1 & 1 & \cdots & 0 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -1 & -1 & \cdots & 1 & 1 \end{pmatrix} \in \mathbb{R}^{n \times n}.$$

Show that Gaussian elimination with partial pivoting can be performed without row interchanges and that the resulting upper triangular factor  $U$  satisfies

$$\max_{i,j} |u_{ij}| = 2^{n-1}.$$

Since

$$\max_{i,j} |(W_n)_{ij}| = 1,$$

deduce that the element growth factor is

$$\rho_n = \frac{\max_{i,j} |u_{ij}|}{\max_{i,j} |(W_n)_{ij}|} = 2^{n-1}.$$

Conclude that the bound

$$\rho_n \leq 2^{n-1}$$

is sharp.